



CYBERSTER

NetSec Internship - Week 5 Task Report

Enterprise-Level Network Architecture & Dynamic Routing

Intern Name:	Muhammad Raza
Instructor Name:	Muhammad Huzaifa Riaz
Company:	CYBERSTER (Batch 2)
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1. Executive Summary

For Week 5, the objective was to design, deploy, and validate a full enterprise-level network architecture using Cisco Packet Tracer. The topology was required to strictly follow a modular block design, incorporating Core, Aggregation, Distribution, and Access blocks. Furthermore, the network had to support a diverse range of endpoint devices - wired PCs and laptops, wireless laptops, and smartphones - via Access Points, and rely entirely on dynamic routing (OSPF) to achieve end-to-end connectivity.

Requirement	How It Was Met
Modular block design	Four distinct blocks implemented: Core, Aggregation, Distribution, and Access, each with dedicated hardware.
Dynamic routing (no statics)	OSPF Area 0 deployed across all Core routers, Aggregation router, and Layer 3 Distribution switches.
Wired endpoint support	Wired PCs and Laptops connected via Cisco 2960 Access Switches in the Access Block.
Wireless endpoint support	Two Wireless Access Points configured with unique SSIDs (VLAN10-WIFI and VLAN20-WIFI) for wireless Laptops and Smartphone.
WAN/External connectivity	Cloud-PT emulation connected to Agg-R1 with a public IP subnet (203.0.113.0/24) to simulate an external WAN boundary.
End-to-end validation	Smartphone0 pinged external gateway 203.0.113.1 - 4/4 packets, 0% loss, 39ms average RTT.

2. Architectural Block Design

The network was segmented into four distinct functional blocks, allowing for scalability, fault isolation, and efficient traffic flow. Each block has a precisely defined role within the enterprise hierarchy.

2.1 Architecture Overview Table

Block	Devices	OSI Layer / Role	Function in This Lab
Core Block	Core-R1, Core-R2 (Cisco 2911)	Layer 3 - High-speed backbone routing	Ultra-fast packet switching between Aggregation and Distribution blocks. No policy overhead.
Aggregation Block	Agg-R1 (Cisco 2911), Cloud-PT	Layer 3 - External WAN/Cloud gateway	Boundary between internal enterprise and external WAN/Internet. Hosts public IP subnet 203.0.113.0/24.
Distribution Block	Dist-SW1, Dist-SW2 (Cisco 3560-24PS)	Layer 3 - Inter-VLAN routing + SVI gateway	Acts as the default gateway for all access-layer subnets. Performs ip routing, creates SVI gateways, and runs OSPF.
Access Block	Access-SW1/2/3 (2960), AP1, AP2, PC0, Laptop0/1/2, Smartphone0	Layer 2 - Endpoint connectivity	Provides wired and wireless access to all end devices. Strictly Layer 2 - no routing performed here.

2.2 Network Topology

The complete enterprise topology is illustrated below with all four blocks visually delineated. Solid lines represent routed point-to-point links; dashed lines represent trunk links.

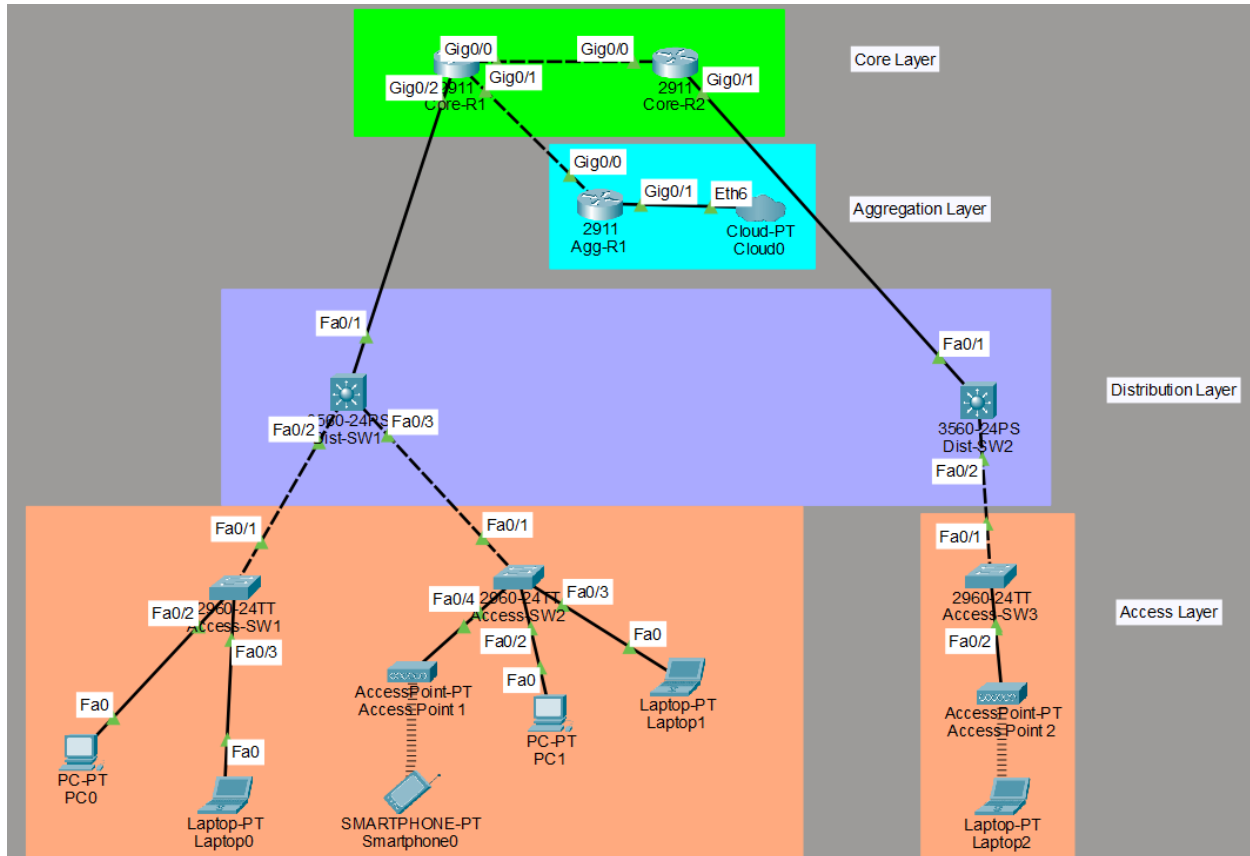


Figure 1: Complete enterprise topology - Core (green), Aggregation (cyan), Distribution (purple), and Access (orange) blocks.

3. Implementation: Configurations & Steps

3.1 Device Placement & Cabling

The physical topology was constructed top-down. The following cable types were used for each connection tier:

Connection	Cable Type	Reason
Core-R1 ↔ Core-R2	Copper Cross-Over (Gigabit)	Router-to-router; same device type requires crossover
Core-R1/R2 ↔ Agg-R1	Copper Cross-Over (Gigabit)	Router-to-router inter-block uplink
Core-R1 ↔ Dist-SW1 (Fa0/1)	Copper Straight-Through	Router-to-switch different device types
Core-R2 ↔ Dist-SW2 (Fa0/1)	Copper Straight-Through	Router-to-switch different device types
Dist-SW1/2 ↔ Access-SW1/2/3	Copper Straight-Through	Switch-to-switch downlink to access tier
Access-SW2 ↔ AP1, SW3 ↔ AP2	Copper Straight-Through	Switch-to-AP standard wired backhaul

3.2 Dynamic Routing - OSPF

To satisfy the requirement of routing across the entire enterprise without static routes, Open Shortest Path First (OSPF) was deployed. router ospf 1 was configured across all Core routers (Core-R1, Core-R2), the Aggregation router (Agg-R1), and both Layer 3 Distribution switches (Dist-SW1, Dist-SW2). All interconnected network segments were advertised into OSPF Area 0, allowing devices to dynamically build and synchronise their routing tables.

OSPF Summary - All Devices Running OSPF Area 0			
Device	Interfaces	Networks Advertised	OSPF Adjacencies Confirmed
Core-R1	Gig0/0, Gig0/1, Gig0/2	10.1.1.0 /8 Area 0	Core-R2 (Gig0/0), Agg-R1 (Gig0/1), Dist-SW1 (Gig0/2)
Core-R2	Gig0/0, Gig0/1	10.1.1.0 /8 Area 0	Core-R1 (Gig0/0), Dist-SW2 (Gig0/1)
Agg-R1	Gig0/0, Gig0/1	10.1.1.0 /8, 203.0.113.0 /24	Core-R1 (Gig0/0), Cloud0 (Gig0/1)

Dist-SW1	Fa0/1 (routed), VLAN 1 SVI	10.1.1.0 /8, 192.168.10.0 /24	Core-R1 (Fa0/1)
Dist-SW2	Fa0/1 (routed), VLAN 1 SVI	10.1.1.0 /8, 192.168.20.0 /24	Core-R2 (Fa0/1)

3.3 Core-R1 Configuration

Core-R1 acts as the primary backbone router, maintaining three Gigabit uplinks: to Core-R2, to Agg-R1, and to Dist-SW1. All three interfaces were assigned point-to-point /30 addresses from the 10.1.1.0/8 block and advertised into OSPF Area 0.

```

Core-R1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Router>
Router>
Router>
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! Link to Core-R2
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Link to Agg-R1
Router(config)#interface gig0/1
Router(config-if)#ip address 10.1.1.5 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Link to Dist-SW1
Router(config)#interface gig0/2
Router(config-if)#ip address 10.1.1.9 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Enable OSPF Routing
Router(config)#router ospf 1
Router(config-router)#network 10.1.1.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
00:50:46: %OSPF-5-ADJCHG: Process 1, Nbr 203.0.113.1 on GigabitEthernet0/1 from LOADING to FULL, Loading Done

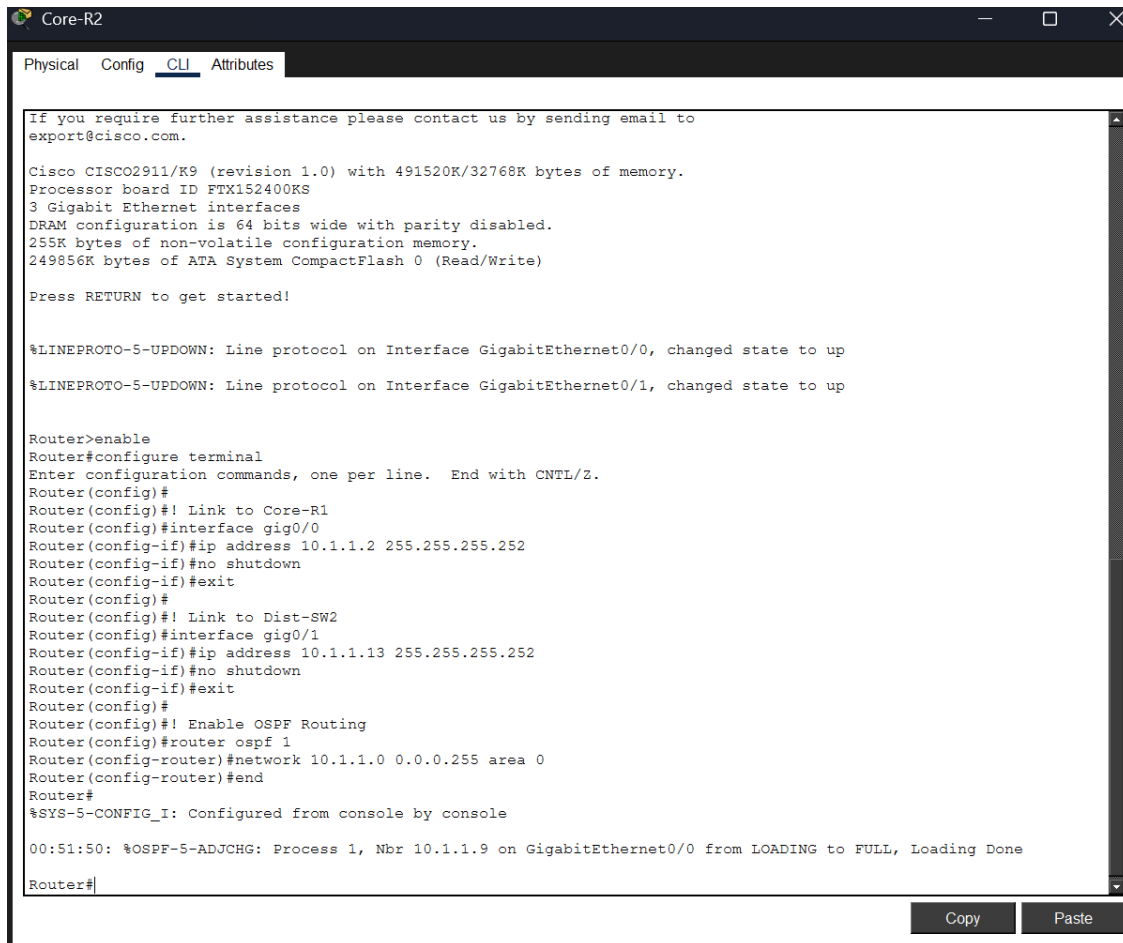
Router#
Copy Paste

```

Figure 2: Core-R1 - Gig0/0 (10.1.1.1) to Core-R2, Gig0/1 (10.1.1.5) to Agg-R1, Gig0/2 (10.1.1.9) to Dist-SW1. OSPF adjacency confirmed with neighbour 203.0.113.1.

3.4 Core-R2 Configuration

Core-R2 provides the second backbone path and connects directly to Dist-SW2, ensuring the right access block has a resilient routed uplink through the Core Layer.



```
Core-R2
Physical Config CLI Attributes

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISCO2911/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
3 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! Link to Core-R1
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Link to Dist-SW2
Router(config)#interface gig0/1
Router(config-if)#ip address 10.1.1.13 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Enable OSPF Routing
Router(config)#router ospf 1
Router(config-router)#network 10.1.1.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

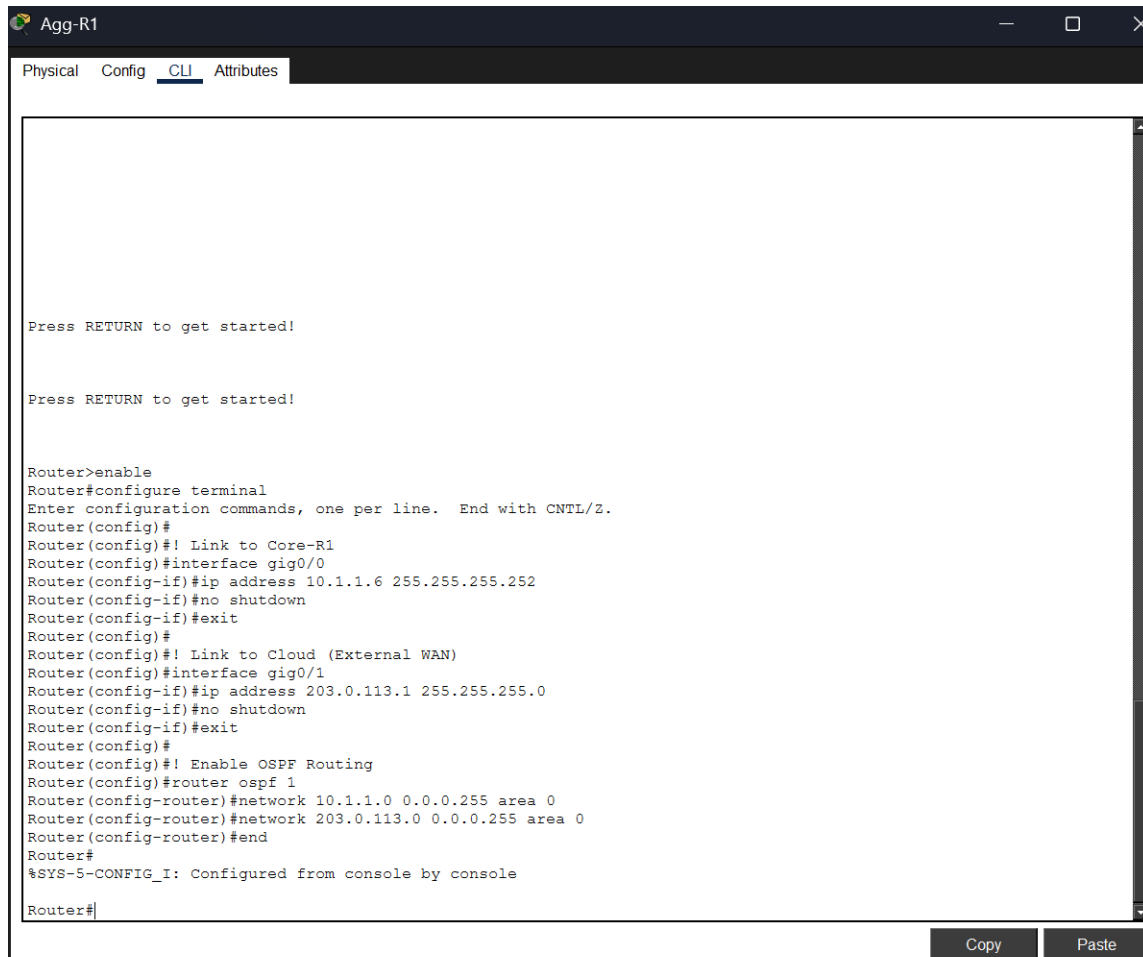
00:51:50: %OSPF-5-ADJCHG: Process 1, Nbr 10.1.1.9 on GigabitEthernet0/0 from LOADING to FULL, Loading Done

Router#
```

Figure 3: Core-R2 - Gig0/0 (10.1.1.2) to Core-R1, Gig0/1 (10.1.1.13) to Dist-SW2. OSPF adjacency confirmed with neighbour 10.1.1.9.

3.5 Agg-R1 Configuration (External Gateway)

Agg-R1 is the boundary router of the Aggregation Block. Its Gig0/0 interface connects inward to Core-R1 using the internal 10.1.1.0/8 space, while its Gig0/1 interface is assigned the public IP 203.0.113.1/24 and connects to Cloud-PT to emulate an external WAN boundary. Both networks are advertised into OSPF, making the external prefix reachable by all internal devices.



```
Agg-R1
Physical Config CLI Attributes

Press RETURN to get started!

Press RETURN to get started!

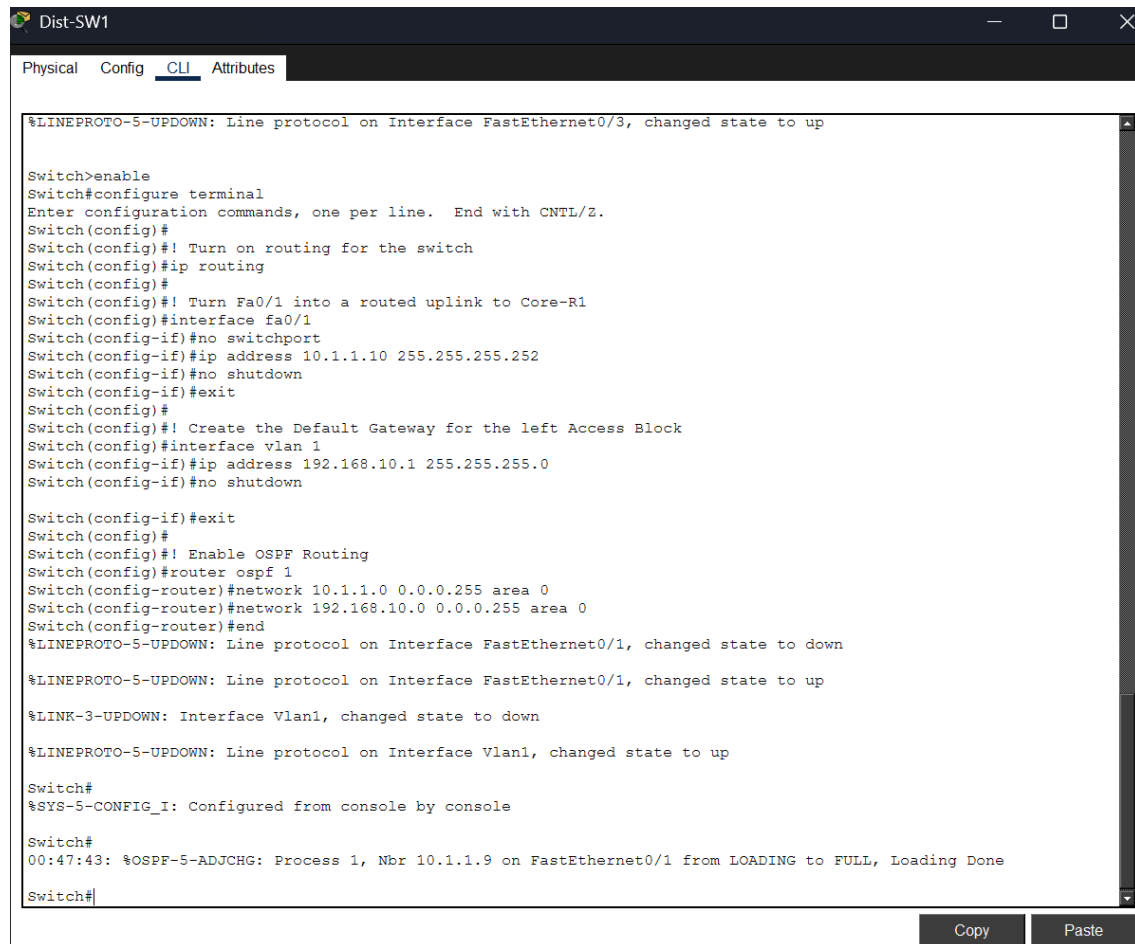
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#
Router(config)#! Link to Core-R1
Router(config)#interface gig0/0
Router(config-if)#ip address 10.1.1.6 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Link to Cloud (External WAN)
Router(config)#interface gig0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#! Enable OSPF Routing
Router(config)#router ospf 1
Router(config-router)#network 10.1.1.0 0.0.0.255 area 0
Router(config-router)#network 203.0.113.0 0.0.0.255 area 0
Router(config-router)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Figure 4: Agg-R1 - Gig0/0 (10.1.1.6) to Core-R1; Gig0/1 (203.0.113.1) to Cloud-PT (WAN boundary). Both networks in OSPF Area 0.

3.6 Dist-SW1 Configuration (Left Access Block)

Dist-SW1 is a Layer 3 Cisco 3560-24PS switch. The ip routing command enables Layer 3 forwarding. Its Fa0/1 interface was converted to a routed port (no switchport) with address 10.1.1.10/30, pointing to Core-R1. A VLAN 1 SVI was created with IP 192.168.10.1/24, which serves as the default gateway for all devices in the left Access Block. Both subnets are advertised into OSPF.



```
Dist-SW1
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#! Turn on routing for the switch
Switch(config)#ip routing
Switch(config)#
Switch(config)#! Turn Fa0/1 into a routed uplink to Core-R1
Switch(config)#interface fa0/1
Switch(config-if)#no switchport
Switch(config-if)#ip address 10.1.1.10 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#! Create the Default Gateway for the left Access Block
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.10.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#exit
Switch(config)#
Switch(config)#! Enable OSPF Routing
Switch(config)#router ospf 1
Switch(config-router)#network 10.1.1.0 0.0.0.255 area 0
Switch(config-router)#network 192.168.10.0 0.0.0.255 area 0
Switch(config-router)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console

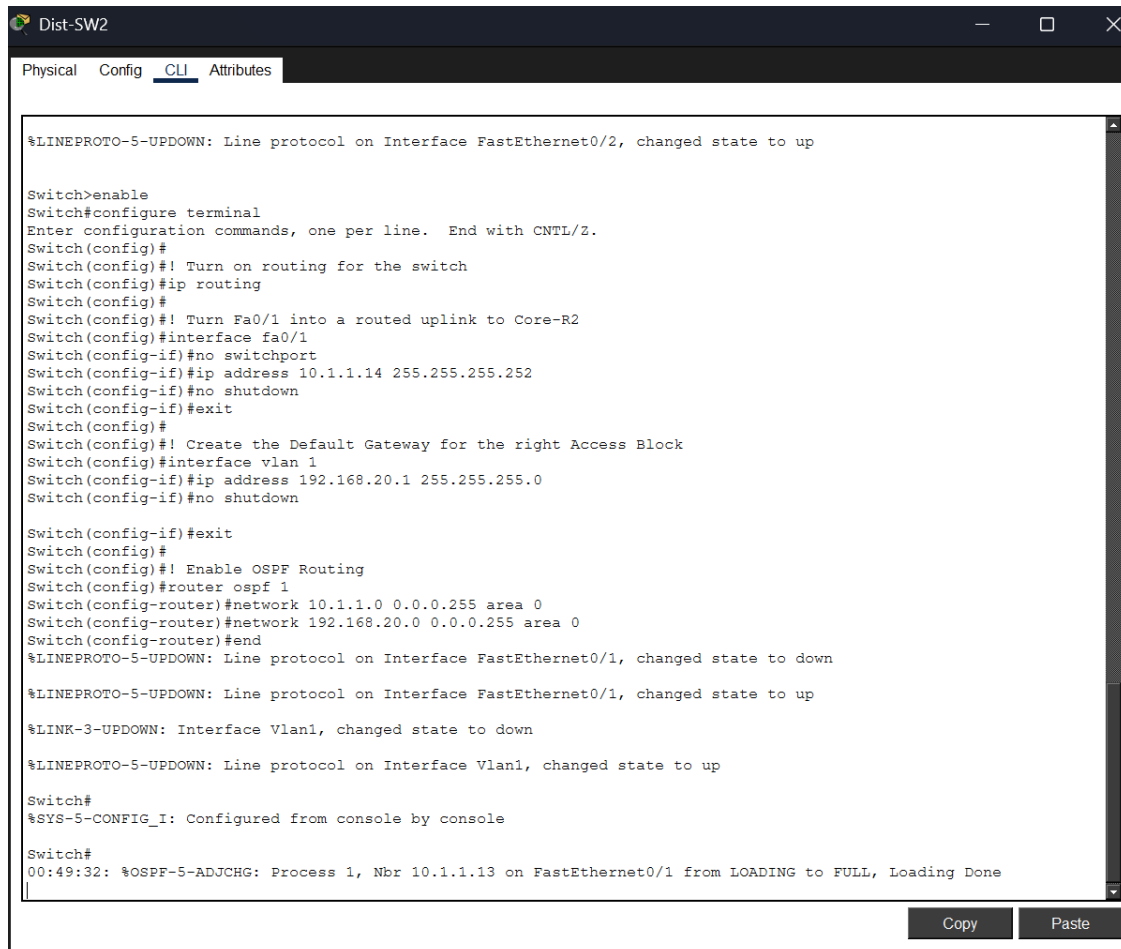
Switch#
00:47:43: %OSPF-5-ADJCHG: Process 1, Nbr 10.1.1.9 on FastEthernet0/1 from LOADING to FULL, Loading Done

Switch#
```

Figure 5: Dist-SW1 - Fa0/1 routed port (10.1.1.10) to Core-R1; VLAN 1 SVI (192.168.10.1) as access-block gateway. OSPF adjacency confirmed.

3.7 Dist-SW2 Configuration (Right Access Block)

Dist-SW2 mirrors the configuration of Dist-SW1 but serves the right Access Block. Its routed uplink to Core-R2 uses 10.1.1.14/30, and its VLAN 1 SVI (192.168.20.1/24) acts as the default gateway for all devices in the right access segment.



```
Dist-SW2
Physical Config CLI Attributes

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up

Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#! Turn on routing for the switch
Switch(config)#ip routing
Switch(config)#
Switch(config)#! Turn Fa0/1 into a routed uplink to Core-R2
Switch(config)#interface fa0/1
Switch(config-if)#no switchport
Switch(config-if)#ip address 10.1.1.14 255.255.255.252
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#! Create the Default Gateway for the right Access Block
Switch(config)#interface vlan 1
Switch(config-if)#ip address 192.168.20.1 255.255.255.0
Switch(config-if)#no shutdown

Switch(config-if)#exit
Switch(config)#
Switch(config)#! Enable OSPF Routing
Switch(config)#router ospf 1
Switch(config-router)#network 10.1.1.0 0.0.0.255 area 0
Switch(config-router)#network 192.168.20.0 0.0.0.255 area 0
Switch(config-router)#end
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

%LINK-3-UPDOWN: Interface Vlan1, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up

Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
00:49:32: %OSPF-5-ADJCHG: Process 1, Nbr 10.1.1.13 on FastEthernet0/1 from LOADING to FULL, Loading Done
```

Figure 6: Dist-SW2 - Fa0/1 routed port (10.1.1.14) to Core-R2; VLAN 1 SVI (192.168.20.1) as access-block gateway. OSPF adjacency confirmed.

3.8 Wireless Access Point Configuration

To manage wireless endpoints within the Access Block, each Access Point was configured with a unique SSID to logically segment wireless traffic and associate devices to the correct subnet. Authentication was set to open (disabled) for this simulation environment.

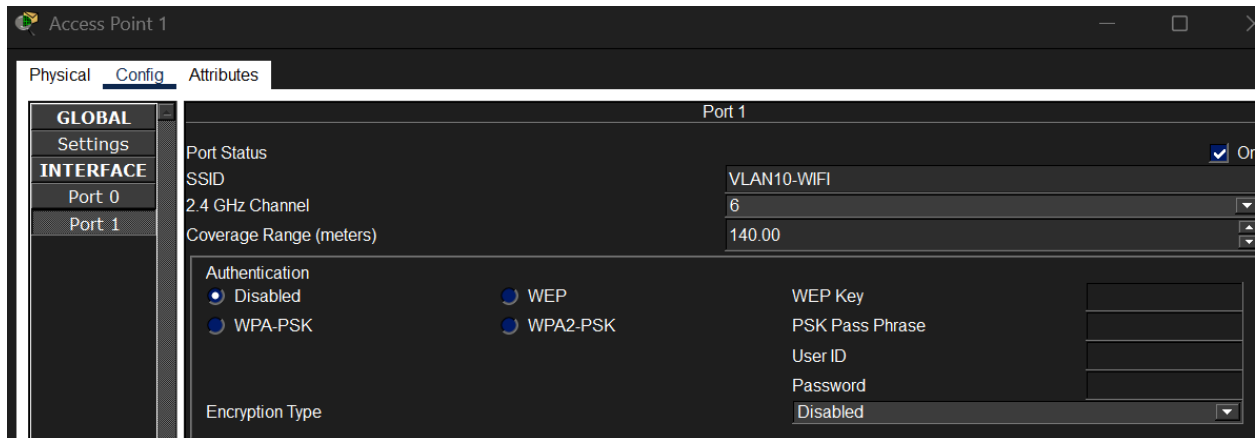


Figure 7: AP1 - SSID: VLAN10-WIFI, Ch. 6, 140m range.

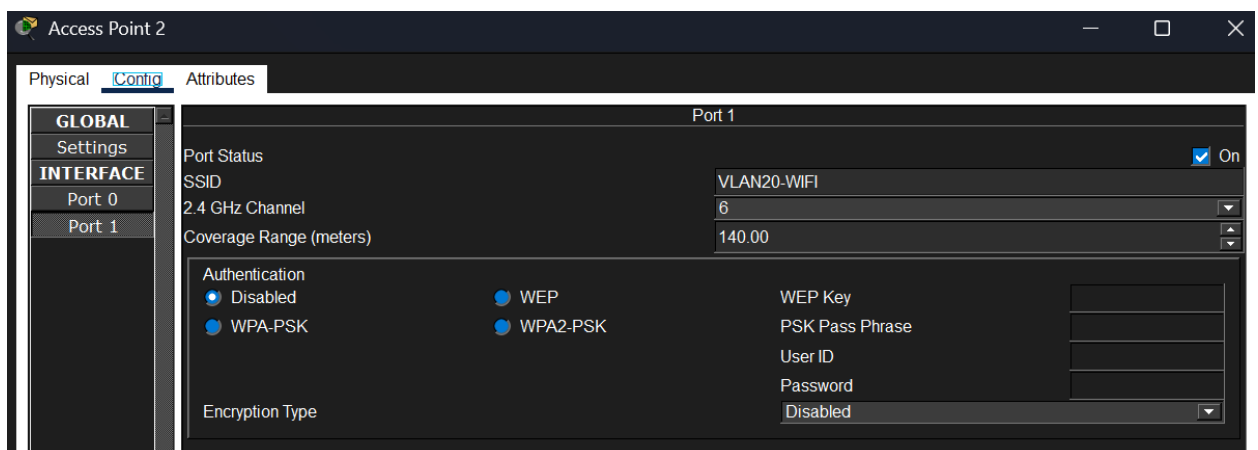


Figure 8: AP2 - SSID: VLAN20-WIFI, Ch. 6, 140m range.

Access Point	SSID	Channel	Connected Wireless Devices
Access Point 1	VLAN10-WIFI	6 (2.4 GHz)	Smartphone0
Access Point 2	VLAN20-WIFI	6 (2.4 GHz)	Laptop2

4. IP Addressing Scheme

The lab uses two distinct IP address spaces: the internal routing backbone (10.1.1.0/8, point-to-point /30 links) and the end-user access subnets (192.168.x.0/24). The external WAN simulation uses public block 203.0.113.0/24.

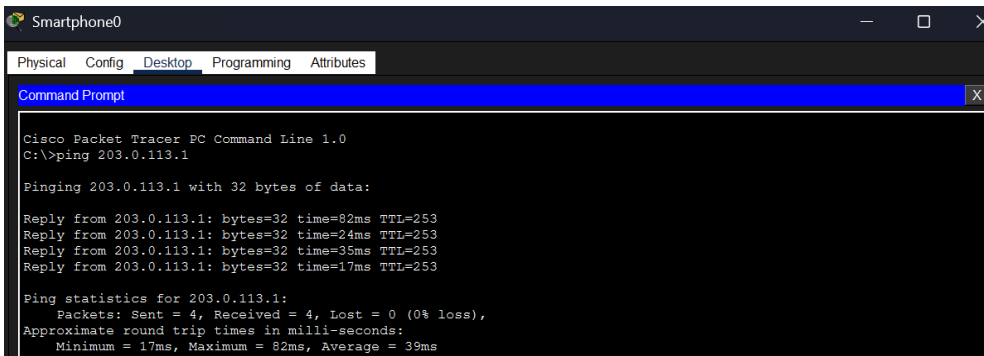
Infrastructure / Backbone Links (Point-to-Point /30)			
Link	Device A Address	Device B Address	Subnet
Core-R1 (Gig0/0) ↔ Core-R2 (Gig0/0)	10.1.1.1 /30	10.1.1.2 /30	10.1.1.0/30
Core-R1 (Gig0/1) ↔ Agg-R1 (Gig0/0)	10.1.1.5 /30	10.1.1.6 /30	10.1.1.4/30
Core-R1 (Gig0/2) ↔ Dist-SW1 (Fa0/1)	10.1.1.9 /30	10.1.1.10 /30	10.1.1.8/30
Core-R2 (Gig0/1) ↔ Dist-SW2 (Fa0/1)	10.1.1.13 /30	10.1.1.14 /30	10.1.1.12/30
Agg-R1 (Gig0/1) ↔ Cloud-PT (WAN)	203.0.113.1 /24	N/A	203.0.113.0/24

End-Device / Access Block Addressing				
Device	Access Switch	IPv4 Address	Default Gateway	Connection
PC0	Access-SW1	192.168.10.10	192.168.10.1 (Dist-SW1)	Wired - Fa0/2
Laptop0	Access-SW1	192.168.10.11	192.168.10.1 (Dist-SW1)	Wired - Fa0/3
Smartphone0	AP1 (SW2)	192.168.10.14	192.168.10.1 (Dist-SW1)	Wireless - VLAN10-WIFI
PC1	Access-SW2	192.168.10.12	192.168.10.1 (Dist-SW1)	Wired - Fa0/2
Laptop1	Access-SW2	192.168.10.13	192.168.10.1 (Dist-SW1)	Wired - Fa0/3
Laptop2	AP2 (SW3)	192.168.20.10	192.168.20.1 (Dist-SW2)	Wireless - VLAN20-WIFI

5. Connectivity Validation

To verify the integrity of the OSPF routing configuration and the full physical topology, an ICMP Echo Request (Ping) was initiated from a wireless edge device - Smartphone0 at the far end of the Access Block - to the external Aggregation gateway interface at 203.0.113.1. This test requires successful packet forwarding across all four network tiers.

Traffic Path	Hop Details
Hop 1 - Access	Smartphone0 (192.168.10.14) → Access Point 1 → Access-SW2 (Layer 2 forwarding)
Hop 2 - Distribution	Access-SW2 → Dist-SW1 (192.168.10.1 SVI gateway, inter-VLAN routing)
Hop 3 - Core	Dist-SW1 → Core-R1 (10.1.1.9 routed link, OSPF-learned route to 203.0.113.0/24)
Hop 4 - Core	Core-R1 → Agg-R1 (10.1.1.5 link, OSPF next-hop selection)
Hop 5 - Aggregation	Agg-R1 → Cloud-PT on 203.0.113.1 (external WAN interface - destination reached)



```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 203.0.113.1

Pinging 203.0.113.1 with 32 bytes of data:

Reply from 203.0.113.1: bytes=32 time=82ms TTL=253
Reply from 203.0.113.1: bytes=32 time=24ms TTL=253
Reply from 203.0.113.1: bytes=32 time=35ms TTL=253
Reply from 203.0.113.1: bytes=32 time=17ms TTL=253

Ping statistics for 203.0.113.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 17ms, Maximum = 82ms, Average = 39ms
```

Figure 9: Smartphone0 → 203.0.113.1 4 packets sent, 4 received, 0% loss. Average RTT: 39ms. TTL=253 confirms 3-hop path through the enterprise tiers.

Source	Destination	Result	TTL / RTT
Smartphone0 - 192.168.10.14	203.0.113.1 (Agg-R1 WAN)	4/4 - 0% Loss	TTL=253 / Avg 39ms

6. Conclusion

Week 5 successfully demonstrated the design, configuration, and validation of a fully functional enterprise-level network architecture in Cisco Packet Tracer. The topology evolved from the three-tier model of Week 4 into a four-block enterprise design, introducing an explicit Aggregation Block to handle external WAN/Cloud boundaries and distributing routing intelligence across the Core, Aggregation, and Distribution tiers.

✓	Achievement
✓	A four-block enterprise architecture (Core, Aggregation, Distribution, Access) was successfully designed and implemented in Cisco Packet Tracer.
✓	OSPF Area 0 was deployed on all five routing devices, eliminating static routes and enabling fully dynamic, self-healing routing table propagation.
✓	The Core Block (Core-R1, Core-R2) provides a redundant backbone ensuring no single point of failure between the Distribution and Aggregation tiers.
✓	The Aggregation Block (Agg-R1 + Cloud-PT) correctly models the enterprise-to-WAN boundary, hosting a public IP subnet (203.0.113.0/24) reachable by all internal endpoints.
✓	The Distribution Block (Dist-SW1, Dist-SW2) converts Layer 3 Cisco 3560 switches into inter-VLAN routers using SVI gateways, bridging the Layer 2 access edge and the Layer 3 core.
✓	Wireless endpoints (Smartphone0, Laptop2) were segmented using dedicated SSIDs (VLAN10-WIFI, VLAN20-WIFI) and successfully associated via their respective Access Points.
✓	End-to-end connectivity was validated: Smartphone0 pinged the external gateway 203.0.113.1 with 4/4 packet success, 0% loss, and an average RTT of 39ms - confirming all four tiers are fully operational.

This Week 5 network represents a production-grade enterprise architecture. The four-block modular design ensures that each tier can scale, be troubleshot, or be upgraded independently - a fundamental property of real-world enterprise networks. The successful OSPF convergence across all devices, culminating in cross-tier reachability from a wireless smartphone at the network edge to a simulated WAN gateway, validates the complete operational integrity of the design.